

Hamiltonian Equipartition as a Convergence Diagnostic for Markov Chains

A Theoretical Report on the Momentum Checker

May 2026

Abstract. We describe and justify a convergence diagnostic for Markov chain Monte Carlo (MCMC) samplers that exploits the factorization of the canonical distribution in phase space. Given a Markov chain targeting the Boltzmann distribution $\pi(x) \propto e^{-\beta V(x)}$, we draw positions from the chain, attach freshly sampled momenta from the canonical momentum distribution, evolve the joint state under Hamiltonian dynamics, and examine the empirical variance of the resulting momenta. By the equipartition theorem, this variance must equal m/β at equilibrium. Any persistent deviation is direct evidence that the position chain has not converged. The diagnostic is global, uses a known exact reference distribution, and naturally admits a one-parameter family of probes indexed by particle mass.

1. Introduction

Markov chain Monte Carlo methods underpin much of modern Bayesian inference and computational statistical mechanics. Their utility is asymptotic: in the limit of infinite samples, ergodic chains produce draws from the desired target distribution. In any finite simulation, however, the sample path carries a bias inherited from its starting state and from the slow modes of the transition kernel. Detecting when this bias has decayed to negligible levels — that is, deciding when a chain has “converged” — is a central practical problem.

Standard diagnostics fall into two broad families. The first compares trajectories of multiple independent chains, the prototype being the Gelman–Rubin $\hat{R}$ statistic [1], which contrasts between-chain and within-chain variance. The second analyzes a single chain through its autocorrelation structure, summarized in the effective sample size [2]. Both families are agnostic to the form of the target: they treat the chain as a generic stationary process and ask whether its empirical moments have stabilized.

The diagnostic developed here is different in kind. It exploits a structural property of distributions of the form $\pi(x) \propto e^{-\beta V(x)}$: when such a distribution is augmented with a Gaussian momentum and evolved under Hamiltonian dynamics, the marginal momentum distribution is preserved exactly. Deviations from the expected momentum variance are therefore unambiguous evidence that the position distribution differs from π . This gives a diagnostic that is global (it senses any disagreement, not just in specific test functions chosen a priori), parametric (the mass introduces a tunable time scale), and grounded in a known closed-form reference.

The rest of the report develops this in detail. Section 2 sets up the canonical distribution and its phase-space factorization. Section 3 reviews the relevant facts about Hamiltonian dynamics. Section 4 specifies the algorithm. Section 5 contains the central theoretical claim and its proof.

Section 6 discusses the mass scan as a multi-scale probe. Sections 7 and 8 compare the diagnostic to standard tools and discuss its limitations.

2. The Canonical Distribution and Its Factorization

Let $V : \mathbb{R}^d \rightarrow \mathbb{R}$ be a potential energy function and let $\beta > 0$ be an inverse temperature. The MCMC target is

$$\pi(x) = \frac{e^{-\beta V(x)}}{Z}, \quad Z = \int_{\mathbb{R}^d} e^{-\beta V(x)} dx, \quad (1)$$

where Z is the normalizing constant. We work in the augmented phase space $\mathbb{R}^d \times \mathbb{R}^d$, introducing an auxiliary momentum variable $p \in \mathbb{R}^d$ and a mass parameter $m > 0$. Define the Hamiltonian

$$H(x, p) = V(x) + \frac{\|p\|^2}{2m}. \quad (2)$$

The associated canonical (Gibbs) distribution on phase space is

$$\rho(x, p) \propto e^{-\beta H(x, p)} = e^{-\beta V(x)} \cdot e^{-\beta \|p\|_{2m}^2}. \quad (3)$$

The decisive observation is that the right-hand side of Equation 3 factorizes. Writing

$$\rho(x, p) = \pi(x) \cdot \mu(p), \quad (4)$$

we identify

$$\mu(p) \propto e^{-\beta \|p\|_{2m}^2} = \prod_{i=1}^d e^{-\beta \frac{p_i^2}{2m}} \quad (5)$$

as a product of independent Gaussians with mean zero and variance m/β in each component. In particular, in the canonical ensemble **position and momentum are statistically independent**, and the marginal momentum distribution is the same regardless of the potential V .

This independence is at the heart of the diagnostic. The momentum has a universal, parameter-free marginal distribution at equilibrium; any chain purporting to sample π can be tested by checking whether the implied joint distribution, when propagated forward in time by Hamiltonian dynamics, respects the universal momentum law.

3. Hamiltonian Dynamics and Preservation of the Canonical Measure

Given the Hamiltonian $H(x, p)$, the equations of motion are

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial H}{\partial x} = -\nabla V(x). \quad (6)$$

Let $\Phi_t : (x_0, p_0) \mapsto (x_t, p_t)$ denote the flow generated by Equation 6 over a time interval t . The flow has two properties of central importance.

Energy conservation. Along any trajectory, Thus $H \circ \Phi_t = H$ for all t .

Volume preservation (Liouville's theorem). The flow is symplectic, hence preserves the $2d$ -dimensional Lebesgue measure on phase space:

Combining these two facts, the canonical density $\rho(x, p) \propto e^{-\beta H(x, p)}$ is invariant under Φ_t . Concretely, if $(X, P) \sim \rho$ and $(X', P') = \Phi_{t(X, P)}$, then $(X', P') \sim \rho$ as well. This is the classical statement that Hamiltonian dynamics preserves the canonical ensemble.

Two corollaries are immediate. First, since $\rho = \pi \cdot \mu$ is the product of two independent marginals, propagation by Φ_t maps the joint law $\pi \times \mu$ to itself. Second, the marginal of ρ over the position coordinate is π , so the position-conditional momentum distribution at any t is again μ . These will be reused below.

4. The Momentum Checker Algorithm

Let $X_1, \dots, X_N$ be position samples produced by a Markov chain targeting π , and let $\hat{\pi}$ denote the empirical distribution of these samples. The algorithm constructs a secondary Markov chain $P_0, P_1, \dots, P_K$ on momentum space as follows.

1. **Initialization.** Draw $P_0 \sim \mu = \mathcal{N}(0, (m/\beta)I_d)$.
2. **Iteration.** For $k = 0, 1, \dots, K-1$:
 1. Draw an index j_k uniformly from $\{1, \dots, N\}$.
 2. Set $X^{(k)} \leftarrow X_{j_k}$ (resample position from $\hat{\pi}$).
 3. Compute $(\tilde{X}, \tilde{P}) = \Phi_{t_k}(X^{(k)}, P_k)$ using a numerical integrator (here, leapfrog over a Poisson-sampled number of steps).
 4. Discard $\tilde{X}$ and set $P_{k+1} \leftarrow \tilde{P}$.
3. **Statistic.** Compute the empirical variance

$$\hat{\sigma}^2 = \frac{1}{K-1} \sum_{k=1}^K (P_k - \bar{P})^2, \quad \bar{P} = \frac{1}{K} \sum_{k=1}^K P_k. \quad (9)$$

Compare $\hat{\sigma}^2$ to the theoretical value m/β .

In words: at each step we replace the position by a fresh draw from $\hat{\pi}$, then let the joint state (X, P) evolve under the Hamiltonian flow, then forget the position. The momentum alone is carried forward. The diagnostic asks whether the long-run distribution of P_k matches the canonical momentum law μ .

5. Why It Detects Non-Convergence

Claim. Suppose $\hat{\pi} = \pi$ exactly. Then $P_k \sim \mu$ for every k , and consequently $\hat{\sigma}^2 \rightarrow m/\beta$ almost surely as $K \rightarrow \infty$.

Proof. By induction on k . At $k = 0$ the claim holds by construction. Suppose $P_k \sim \mu$. In step (ii), $X^{(k)}$ is an independent draw from $\hat{\pi} = \pi$. Therefore $(X^{(k)}, P_k)$ has joint law $\pi \times \mu = \rho$, the canonical distribution on phase space. By the invariance result of Section 3, $(\tilde{X}, \tilde{P}) = \Phi_{t_k}(X^{(k)}, P_k)$ also has law ρ . The marginal of ρ in the momentum coordinate is μ , so $P_{k+1} = \tilde{P} \sim \mu$. The strong law of large numbers then gives $\hat{\sigma}^2 \rightarrow \text{Var}(\mu) = m/\beta$. $\square$

This is the positive half of the diagnostic. The negative half — that $\hat{\sigma}^2 \neq m/\beta$ when $\hat{\pi} \neq \pi$ — is the operational claim on which the diagnostic depends, and we treat it next.

Why disagreement shows up. Suppose now that $\hat{\pi}$ is some distribution $\tilde{\pi} \neq \pi$. At step k , the joint law of $(X^{(k)}, P_k)$ is $\tilde{\pi} \times \mu$, which is **not** invariant under Φ_t unless $\tilde{\pi} \propto e^{-\beta V}$. Energy conservation along a trajectory gives so the post-evolution kinetic energy is Taking expectations under the joint law $\tilde{\pi}(X^{(k)}) \times \mu(P_k)$, Under the **true** equilibrium $\tilde{\pi} = \pi$, the two potential terms cancel because the law of $\tilde{X}$ marginalizes to π . Under any other $\tilde{\pi}$, the cancellation fails, and the kinetic energy systematically drifts. Iterating, the stationary kinetic energy of the P_k chain deviates from its canonical value of $d/(2\beta)$. Equivalently, the per-coordinate variance of P_k deviates from m/β .

The diagnostic is therefore not detecting some subtle distributional feature: it is detecting a violation of equipartition. This makes it both interpretable and sharp.

6. The Mass Scan as a Multi-Scale Probe

The diagnostic depends on a free parameter, the mass m . Two different choices probe the same chain on different physical time scales, and a scan over m provides a richer view than any single value.

The natural velocity scale of a particle at temperature β^{-1} and mass m is

$$v_{\text{rms}} = \sqrt{\langle \|\frac{p}{m}\|^2 \rangle} = \sqrt{\frac{d}{m\beta}}, \quad (13)$$

so light particles move quickly and heavy particles move slowly. Over a fixed simulation time τ , the lightest particle in a scan covers a distance proportional to $\tau/\sqrt{m}$.

This has two important consequences for the diagnostic:

1. **Light masses probe global structure.** A small m means the particle sweeps through a large region of position space during τ . The momentum at the end of the trajectory is sensitive to the full mass of position samples encountered along the way, including regions $\hat{\pi}$ may have failed to populate correctly.
2. **Heavy masses probe local structure.** Large m means the particle barely moves during τ . The diagnostic then sees only the local potential curvature near $X^{(k)}$, and is less sensitive to long-range distributional errors.

A scan plot of $\hat{\sigma}(m)$ against m thus has interpretive value: the expected curve is $\sqrt{m/\beta}$, growing as $\sqrt{m}$, and systematic deviations at small m versus large m have different physical meanings.

A second, more practical concern is that small m couples to numerical stability of the leapfrog integrator. The position update is $x \leftarrow x + (\varepsilon/m)p$, so for fixed ε the per-step displacement grows as m^{-1} . In the diagnostic code this is mitigated by scaling ε linearly with m , keeping the ratio ε/m — and hence the numerical stability of the leapfrog — constant across the mass sweep.

7. Relation to Standard Diagnostics

It is worth situating the momentum checker among other tools.

Gelman–Rubin $\hat{R}$. This contrasts variances within and between multiple chains. It requires running parallel chains from overdispersed starting points, and detects failures of mixing between modes that no single chain has crossed. The momentum checker uses a single chain (the position chain may be one HMC, one Metropolis–Hastings, or a concatenation of several) and detects failures of the **marginal** distribution to equal π . The two diagnostics are not redundant: $\hat{R}$ can pass while the momentum check fails (the chains agree on a wrong distribution) and vice versa (the chains disagree but each marginally matches π — unusual but possible).

Effective sample size and integrated autocorrelation. These estimate the variance reduction of empirical averages over independent samples. They are silent about whether the underlying mean is correct; a chain stuck in one mode of a multimodal target may report high ESS and still be wrong. The momentum checker, in contrast, fails systematically if the mode weights are wrong, because the implied potential-energy expectation used in Equation 11 is incorrect.

Test-function diagnostics. One can choose specific observables $f_1, \dots, f_J$ and check whether $\mathbb{E}_{\hat{\pi}}(f_j)$ matches some known value. The momentum checker can be seen as a specific instance of this strategy, with f implicitly defined by the Hamiltonian flow. The advantage is that no test function need be selected a priori: the dynamics itself integrates the position information into a single scalar (the momentum variance) whose target value is known exactly from physics.

8. Limitations

Discretization bias. The Hamiltonian flow Φ_t is approximated by the leapfrog integrator. Leapfrog is symplectic but not exact: it conserves a **modified** Hamiltonian $\tilde{H} = H + O(\varepsilon^2)$, not H itself. The invariance argument therefore applies to the modified canonical density $\exp(-\beta\tilde{H})$, not $\exp(-\beta H)$. The difference is $O(\varepsilon^2)$ for leapfrog and decays as $\varepsilon \rightarrow 0$; in practice the diagnostic must use ε small enough that this bias is below the precision of interest.

Finite-sample noise. The empirical variance $\hat{\sigma}^2$ has its own sampling distribution. For a Gaussian sample of size K , the standard error of the variance estimator scales as $\sqrt{\frac{2}{K}} \cdot \sigma^2$, so $K = O(10^3)$ suffices for percent-level precision in benign cases but may be inadequate for detecting subtle bias. Burn-in must also be discarded before computing $\hat{\sigma}^2$, and the choice of cutoff affects sensitivity.

Multimodal targets. If $\hat{\pi}$ visits only one of several modes, the potential-energy expectation $\mathbb{E}_{\hat{\pi}}(V)$ may still happen to match its true value if the mode is symmetric or representative. The diagnostic then misses the failure. Combining the momentum check with multi-chain comparisons (Gelman–Rubin) closes this gap.

Cost. Each diagnostic iteration requires a leapfrog trajectory of length comparable to a full HMC step, so the diagnostic is roughly as expensive as generating one HMC sample per check iteration. A scan over J masses with K iterations each multiplies this cost by JK . For typical settings ($J \approx 30$, $K \approx 10^3$) the overhead is comparable to running the original chain itself.

9. Conclusion

The momentum checker is a diagnostic with an unusually transparent theoretical basis: it tests the equipartition theorem, a closed-form prediction of statistical mechanics, against a Hamiltonian propagation of the chain’s empirical position distribution. Convergence of the chain implies the test passes; failure of the test implies the chain has not converged.

What makes the diagnostic particularly informative in the MCMC setting is that the reference distribution against which the chain is compared is **universal** — the same Gaussian regardless of V — and **exact** — known in closed form independent of any numerics. Both properties are rare in convergence diagnostics, which typically rely either on comparisons among independent chains (whose mutual agreement is not the same as agreement with π) or on stationarity of summary statistics over time (which says nothing about the value to which they have stationarized). By contrast, the expected value m/β has a derivation a few lines long, and any sustained deviation is direct evidence of bias.

The mass sweep adds a second axis of diagnosis: rather than producing a single number, it produces a curve, and the shape of that curve carries interpretive information about whether the chain has failed at small or large scales. Together with conventional tools — $\hat{R}$, autocorrelation, trace plots — the momentum checker provides a physically motivated, distribution-aware complement that is especially natural in settings where HMC is already being used and the Hamiltonian machinery is already implemented.

References

- [1] A. Gelman and D. B. Rubin. *Inference from iterative simulation using multiple sequences*. Statistical Science, 7(4):457–472, 1992.
- [2] C. J. Geyer. *Practical Markov chain Monte Carlo*. Statistical Science, 7(4):473–483, 1992.
- [3] R. M. Neal. *MCMC using Hamiltonian dynamics*. In *Handbook of Markov Chain Monte Carlo*, pages 113–162. Chapman & Hall/CRC, 2011.
- [4] M. Betancourt. *A conceptual introduction to Hamiltonian Monte Carlo*. arXiv:1701.02434, 2017.
- [5] L. D. Landau and E. M. Lifshitz. *Statistical Physics, Part 1*, 3rd edition. Pergamon Press, 1980.